One Policy, Infinite NPCs: A Vision for Scalable Persona-Conditioned NPC Control in Life Simulation Games

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***Abstract*—Life simulation games require hundreds to thousands of non-player characters (NPCs) that each behave consistently with a distinct personality across an entire play session. Current approaches—hand-authored behavior trees, per-character RL policies, unsupervised skill discovery, and LLM-as-policy—each fail on one or more of the four axes that matter for practical deployment: persona consistency, natural-language controllability, zero-shot generalization, and real-time inference. We argue that *persona-conditioned shared policies*—a single RL policy conditioned on frozen LLM embeddings of free-form persona descriptions—represent a promising architectural direction that can satisfy all four axes simultaneously. To support this argument we introduce PCSP (Persona-Conditioned Shared Policy), an early proof-of-concept combining frozen LLM encoding, LoRA projection, FiLM conditioning, and a PPO + InfoNCE consistency + KL diversity co-training objective. On a 300-persona life-simulation benchmark, PCSP achieves zero-shot persona identification 11× above chance, Spearman $\rho = 0.73$ semantic-behavioral alignment, and 22× faster inference than LLM-as-policy. We then lay out a concrete research agenda—covering dynamic personas, social emergence, long-horizon memory, richer environments, human evaluation, and authoring tools—for the community to build on this direction.**

***Index Terms*—game AI, NPC personalization, reinforcement learning, persona conditioning, large language models, life simulation**

I. Introduction

Modern life simulation games—*The Sims*, *Animal Crossing*, *inZOI*, and emerging open-world titles—place NPCs at the center of the player experience. For these games to feel alive, each NPC must behave consistently with a distinct personality: the gregarious chef and the reclusive artist should pursue the same biological needs through recognizably different activity patterns, social approaches, and daily rhythms. At scale—hundreds to thousands of NPCs per game world—this creates a fundamental challenge that current game AI architectures were not designed to address.

A. The NPC Personalization Scaling Gap

a) Behavior trees: remain the industrial standard [1]. A skilled designer hand-crafts decision logic for each character archetype. This produces believable, predictable NPCs, but authoring cost scales linearly with character count, and trees are brittle outside authored scenarios. Ten thousand distinct NPC personalities require ten thousand hand-crafted trees.

b) Per-NPC RL: appears to offer automation: train a separate policy for each persona. In principle this allows unlimited characters with arbitrary behaviors. In practice, memory and training cost also scale linearly, and each new persona requires a full retraining cycle—a prohibitive expense for live-service games that regularly introduce new characters.

c) LLM-as-policy: methods [2]–[4] achieve rich, natural-language-grounded persona expression by querying a language model at every decision step. This works well in offline or turn-based contexts, but current LLMs require 43–500 ms per inference call [5], which exceeds the 16–33 ms per-frame budget of real-time game simulations. Generative Agents [2] sidestep this by operating on minute-resolution plans rather than per-frame actions, which suits narrative simulation but cannot drive moment-to-moment NPC movement and activity selection.

d) Unsupervised skill discovery: (DIAYN [6], CIC [7]) learns a shared policy conditioned on latent skill codes, achieving behavioral diversity without per-character training. But the codes carry no semantic content. A game designer cannot specify "this NPC should be agreeable and fitness-oriented" by selecting a latent index. Natural-language controllability is not part of the design.

B. A Promising Architectural Direction

We argue that the solution lies in a different decomposition: *compute a rich persona representation once per NPC, then condition a single lightweight policy on that representation at every step.*

The key enabler is the modern LLM embedding space. A frozen language model maps free-form persona descriptions to dense vectors that capture semantic relationships among personalities. A shared policy conditioned on these embeddings can generalize to *any* persona a designer writes, without retraining, because the continuous embedding space provides coverage of the entire personality manifold. At inference time, the LLM is called once per NPC lifetime; the policy network—at most a few hundred kilobytes—runs at full game speed.

TABLE I
PARADIGM COMPARISON. ✓ = SATISFIED, × = NOT SATISFIED, △ = PARTIALLY SATISFIED.

Paradigm	Persona consist.	NL control.	Zero-shot gen.	Real-time (<5 ms)
Behavior trees	△	×	×	✓
Goal-cond. RL	×	×	×	✓
Lang-cond. RL	△	✓	△	✓
Skill discovery	△	×	×	✓
Per-NPC RL	✓	×	×	✓
LLM-as-policy	✓	✓	✓	×
PCSP (ours)	✓	✓	✓	✓

This architectural direction, which we call *persona-conditioned shared policies*, is not yet mature. The design space of embedding methods, conditioning architectures, training objectives, and evaluation protocols remains largely unexplored. The goal of this paper is threefold: (1) articulate why this direction is worth pursuing, (2) present an early concrete instance (PCSP) as an existence proof, and (3) identify the research questions that must be answered to make the approach viable at game production scale.

C. Paper Contributions

1) **Research agenda**: We identify persona-conditioned shared policies as an underexplored architectural direction and lay out six concrete open problems (§V).
2) **PCSP**: An early proof-of-concept combining frozen Qwen3-0.6B-Embedding, LoRA projection, FiLM conditioning, and a PPO + InfoNCE + KL co-training objective (§III).
3) **Initial evidence**: On a 300-persona life-simulation benchmark, PCSP achieves zero-shot persona identification 11× above chance, Spearman $\rho = 0.73$ semantic-behavioral alignment, and 22× faster inference than LLM-as-policy (§IV).
4) **Evaluation agenda**: A proposal for metrics and benchmarks suited to persona fidelity evaluation beyond task reward (§VI).

II. WHY CURRENT PARADIGMS FALL SHORT

Table I evaluates six paradigms against the four axes that matter for practical life-simulation NPC deployment.

a) Goal-conditioned RL.: UVFA [8] and HER [9] condition policies on goal *states*—what to achieve. Persona describes *how to behave*: two NPCs with identical needs but different personalities should fulfill them through different activity sequences. Goal conditioning addresses the wrong axis of variation and provides no natural-language interface.

b) Language-conditioned RL.: BabyAI [10] conditions on natural-language *instructions* that change each episode. Persona is a stable trait persisting over the NPC's lifetime; treating it as an episode-level instruction discards the long-horizon consistency constraint. Decision Transformer [11] and similar sequence-modeling approaches condition behavior on past returns or goals, but not on static identity descriptors that must generalize zero-shot.

c) Unsupervised skill discovery.: DIAYN [6] and CIC [7] achieve behavioral diversity via learned latent codes, but the codes carry no semantic content. Generalizing to a new designer-specified persona requires solving an inverse problem: find the latent code for "an introverted accountant who prioritizes learning." This is not supported by design.

d) LLM-as-policy.: Generative Agents [2], Voyager [3], and ReAct [4] produce rich, persona-consistent behavior but at 43–500 ms per step. Techniques such as action caching, plan reuse, and smaller distilled models can reduce latency, but the fundamental tension—rich language reasoning versus real-time step budgets—remains. Generalist agents such as Gato [12] demonstrate that a single model can handle diverse tasks, but inference cost is similar.

e) Summary.: No existing paradigm satisfies all four axes. The design space between "fast but no NL control" and "NL control but too slow" has not been systematically explored. Persona-conditioned shared policies are a concrete proposal for how to close this gap.

III. PCSP: A CONCRETE EARLY INSTANCE

We present PCSP as one specific point in the design space of persona-conditioned shared policies. The components we have chosen are motivated by ablation evidence but are not canonical; they represent a tractable first step, not a final architecture.

A. Environment: Mini-Inzoi

Mini-Inzoi is a PettingZoo AEC life-simulation environment on a 6×6 grid with 4 agents, 8 bio-social needs (hunger, sleep, social, leisure, hygiene, fitness, work, learning), and 12 discrete actions (8 activity + 4 movement). Observation $s \in \mathbb{R}^{20}$: position (2), time-of-day (1), needs (8), and summaries of the other three agents (9). Reward combines needs satisfaction, persona-aligned activity bonuses (+0.5 for preferred actions), and social bonuses scaled by Big Five compatibility [13] ($0.2 + 0.3 \times \text{cos-sim}(\text{BF}_i, \text{BF}_j)$).

Persona dataset. 300 persona texts are generated from 15 Big Five archetypes × 20 occupations (240 train / 60 zero-shot test). Each text is a 2–3 sentence natural-language description that an NPC designer might write for a new character.

B. Persona Encoding

Given persona text p, we compute a frozen LLM embedding $\hat{e}_p \in \mathbb{R}^{1024}$ using Qwen3-0.6B-Embedding [14] (last-token pooling, L2-normalized). This is computed once per NPC lifetime (14.6 ms/persona), imposing negligible runtime cost.

A learned LoRA projection [15] ($r = 16$, $\text{lr} = 10^{-4}$) maps $\hat{e}_p \to e_p \in \mathbb{R}^{64}$:

$$e_p = W_{\text{down}}^{\top}\, \sigma\big(W_{\text{up}}^{\top} \hat{e}_p\big)\,, \quad W_{\text{up}} \in \mathbb{R}^{1024\times512},\ W_{\text{down}} \in \mathbb{R}^{512\times64}.$$

The projection is necessary: raw Qwen3 embeddings over-represent occupational similarity relative to personality similarity (salesperson↔executive: 0.61 vs. trainer↔blogger: 0.33).

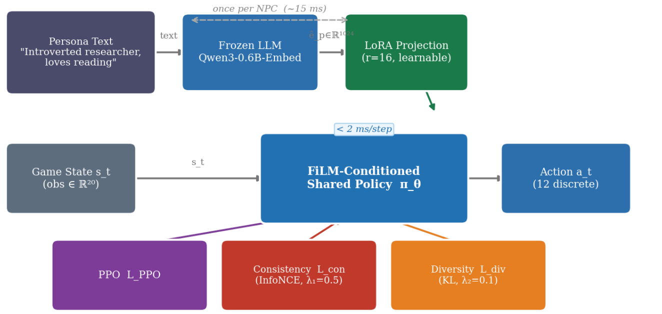

Fig. 1. **PCSP system overview.** A frozen LLM encoder computes a once-per-NPC persona embedding. A learned LoRA projection adapts it to behavior space. The shared FiLM-conditioned policy runs at game speed (<2 ms/step) and is trained with PPO + consistency + diversity co-objectives.

After LoRA training, Spearman ρ between embedding distance and behavioral KL increases from 0.384 to 0.728 (§IV).

C. FiLM-Conditioned Shared Policy

The shared policy $\pi_\theta(a \mid s, e_p)$ is a 3-layer MLP (256-256-10) where each hidden layer h_ℓ is conditioned via FiLM [16]:

$$h'_\ell = \gamma_\ell(e_p) \odot h_\ell + \beta_\ell(e_p),$$

with small linear networks $\gamma_\ell, \beta_\ell : \mathbb{R}^{64} \to \mathbb{R}^{256}$. FiLM injects persona information at *every* hidden layer, preventing the persona signal from being diluted in deep representations. An identical FiLM-conditioned value network $V_\phi(s, e_p)$ serves as the critic. Total trainable parameters: 207K (excluding the frozen LLM).

D. Co-Training Objective

$$\mathcal{L} = \mathcal{L}_{\text{PPO}} + \lambda_1 \mathcal{L}_{\text{consist}} + \lambda_2 \mathcal{L}_{\text{diverse}}, \qquad (1)$$

with $\lambda_1 = 0.5$, $\lambda_2 = 0.1$ (PPO: $\gamma = 0.99$, $\epsilon = 0.2$, GAE-$\lambda = 0.95$, batch$\,=2048$ [17]).

Consistency loss. A 2-layer GRU trajectory encoder $g_\phi(\tau) \in \mathbb{R}^{64}$ maps episode trajectories to L2-normalized embeddings. InfoNCE [18] ($T = 0.07$) aligns trajectory embeddings with their originating persona embeddings:

$$\mathcal{L}_{\text{consist}} = -\log \frac{\exp\big(\text{sim}(g_\phi(\tau), e_p)/T\big)}{\sum_{p' \in \mathcal{B}} \exp\big(\text{sim}(g_\phi(\tau), e_{p'})/T\big)}. \qquad (2)$$

This prevents the policy from producing trajectories that cannot be traced back to their conditioning persona.

Diversity loss. We maximize the expected KL divergence between policy distributions induced by different personas at shared states:

$$\mathcal{L}_{\text{diverse}} = -\mathbb{E}_{s,\, p \neq p'}\Big[D_{\text{KL}}\big(\pi_\theta(\cdot \mid s, e_p) \,\|\, \pi_\theta(\cdot \mid s, e_{p'})\big)\Big], \quad (3)$$

computed as a batched forward pass over $n = 8$ sampled personas at 32 states. This discourages behavioral collapse toward a persona-agnostic average.

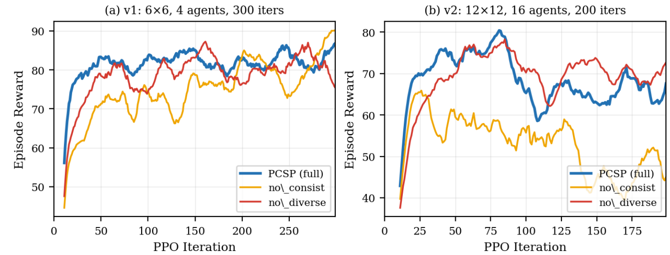

Fig. 2. **Training reward curves at two scales.** (a) v1 (6×6, 4 agents, 300 iters). (b) v2 (12×12, 16 agents, 200 iters). The diversity loss (no_diverse) is the most critical ablation at both scales: its removal causes the largest reward drop and, more importantly, behavioral collapse (KL: v1 0.39, v2 0.48). The consistency loss (no_consist) trades slightly higher reward for total loss of persona traceability.

E. What This Is Not

PCSP is a laboratory-scale proof-of-concept, not a production-ready NPC system. The two Mini-Inzoi variants (6×6 and 12×12) are intentionally minimal; the personas are synthetically generated; and human evaluation has not been completed. We present these results as evidence that the architectural direction is viable, not as a claim that NPC personalization has been solved.

IV. Initial Evidence

We report experiments at two scales to test both the base design choices and whether the architectural properties hold as the environment grows.

A. Experimental Setup

v1 (base scale). Mini-Inzoi 6×6, 4 agents, 300 personas (240 train / 60 zero-shot test). All models train for 300 PPO iterations (≈1.9M steps) on an NVIDIA RTX 6000 Ada (49 GB VRAM). Baselines: **B1** No-Persona PPO; **B3** SBERT-conditioned policy (`all-MiniLM-L6-v2`, 384-dim, frozen); **B4** DIAYN (random 64-dim); **B5** LLM-as-policy (Qwen3-1.7B).

v2 (expanded scale). Mini-Inzoi **12×12, 16 agents, 500 personas** (400 train / 100 zero-shot test), 200 PPO iterations. Same hyperparameters and evaluation protocol as v1. The observation dimension expands to 56 (position 2, time 1, needs 8, 15 other agents × 3). Random zero-shot chance is 1.0% (1/100 test personas).

B. Results

Tables II and III show key metrics at each scale. Figure 2 shows training reward curves at both scales. Full ablation details appear in the supplementary.

C. Key Observations

Three findings from the v1 ablation study are replicated in v2, supporting their generality across environment scales.

Consistency loss is necessary for persona traceability. In v1, removing it drops zero-shot accuracy from 19.3% to 1.7% (random). In v2, it drops from 2.3% to exactly 0% against

TABLE II

V1 RESULTS (6×6, 4 AGENTS, 300 PERSONAS). ZS ACC: ZERO-SHOT k-NN ACCURACY ON 60 UNSEEN PERSONAS (RANDOM $\approx$1.7%). ρ: SPEARMAN (EMBEDDING DIST. VS. KL).

Model	Reward ↑	ZS Acc ↑	ρ ↑	Lat. ↓
PCSP (full)	83.4	**0.193**	**0.728**	1.97 ms
PCSP (no_consist)	84.3	0.017*	0.638	2.03 ms
PCSP (no_diverse)	76.8	0.147	—†	1.79 ms
B1 No-Persona	73.2	—	—	1.83 ms
B3 SBERT	**86.2**	—	—	1.88 ms
B5 LLM-policy	—	—	—	43.7 ms

* Near-random (failure mode). †KL $\approx$0.39; ρ not meaningful.

TABLE III

V2 RESULTS (12×12, 16 AGENTS, 500 PERSONAS). ZS ACC ON 100 UNSEEN PERSONAS (RANDOM $=$1.0%). LOWER ABSOLUTE REWARD REFLECTS THE HARDER, LARGER ENVIRONMENT.

Model	Reward ↑	ZS Acc ↑	ρ ↑	KL ↑
PCSP (full)	64.2	**0.023**	**0.725**	**5.40**
PCSP (no_consist)	50.1	0.000*	0.253	16.47‡
PCSP (no_diverse)	71.6	0.013	—†	0.48
PCSP (concat)	**74.8**	0.027	—†	1.34

* Exactly 0 (total failure). †KL too low; ρ not meaningful. ‡High KL without consistency reflects unstructured divergence, not persona alignment.

a 1.0% random baseline. In both cases, trajectories carry no recoverable persona signal without this loss.

Diversity loss prevents behavioral collapse. In v1, removing it reduces mean KL from 5.87 to 0.39 ($15\times$). In v2, from 5.40 to 0.48 ($11\times$). The collapse pattern is consistent: without this loss, the policy converges to a persona-agnostic average regardless of environment size.

Spearman ρ is preserved across scales. PCSP (full) achieves ρ=0.728 in v1 and ρ=0.725 in v2—a difference of 0.003 despite a $4\times$ increase in grid area, $4\times$ more agents, and 67% more personas. This suggests that semantic-behavioral alignment is a stable property of the architecture, not an artefact of the minimal environment.

D. Significance

The absolute task reward and zero-shot accuracy are lower in v2, as expected for a substantially harder environment with more test personas. The key point is not the absolute numbers but the *replication of qualitative structure*: the same architectural decisions matter at both scales, and the semantic-behavioral alignment property ($\rho \approx 0.73$) is preserved. Figure 3 shows the zero-shot distribution for v1; the v2 pattern is qualitatively similar with narrower per-persona bars. Figure 4 shows that the monotone relationship between persona embedding distance and behavioral KL is virtually unchanged across the two environments (ρ=0.728 vs. 0.725), providing the clearest evidence that semantic-behavioral alignment is a stable property of the architecture. These results are *early encouraging signs* that the direction is worth pursuing at larger scale.

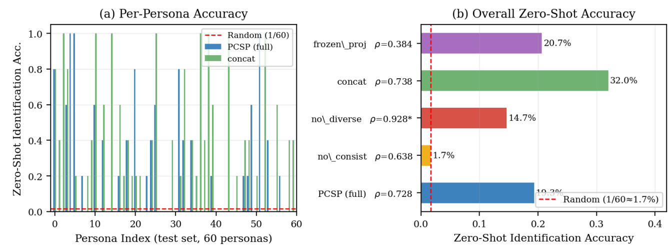

Fig. 3. **Zero-shot generalization on 60 unseen personas (v1).** PCSP (full) achieves 19.3% trajectory-to-persona identification; the red dashed line marks random chance (1.7%). Removing the consistency loss collapses accuracy to 1.7%. The v2 experiment (100 unseen personas) replicates this qualitative pattern.

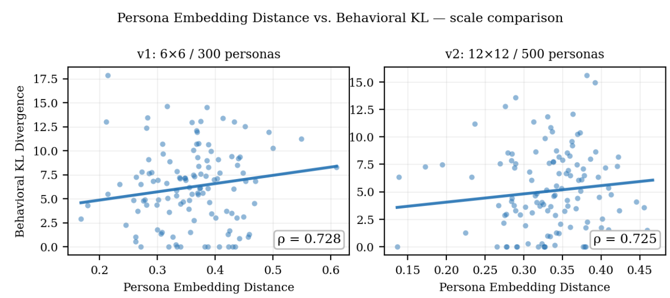

Fig. 4. **Persona embedding distance vs. behavioral KL divergence at two scales (PCSP full).** Left: v1 (ρ=0.728). Right: v2 (ρ=0.725). The monotone alignment between LLM semantic space and behavior space is preserved when moving to a $4\times$ larger grid with $4\times$ more agents and 67% more personas.

V. RESEARCH AGENDA FOR INFINITE NPCS

The results above are encouraging, but Mini-Inzoi is a deliberately minimal environment. Reaching production-quality NPC personalization requires answering six open research questions. We outline each with the core challenge and candidate approaches.

A. Dynamic Personas and Identity Evolution

Current PCSP encodes a static persona text computed once at NPC creation. Real characters evolve: a traumatic in-game event should shift an NPC toward neuroticism; forming a friendship should increase agreeableness over time.

The key challenge is *representation*: how should an evolving personality be maintained and encoded without expensive LLM calls at every update? Candidate approaches include (a) maintaining a low-dimensional latent personality state updated by a lightweight delta model at milestone events, (b) conditioning the LoRA projection on recent interaction history via a rolling context encoder, or (c) hierarchical persona representations that separate stable traits (*e.g.* conscientiousness) from volatile moods (*e.g.* current frustration level).

B. Socially Emergent Multi-Agent Behavior

In multi-agent settings, personas interact: an introverted NPC paired with an extroverted one may develop avoidance patterns; two agreeable NPCs may form stable dyads. These emergent social structures are a major source of narrative interest in life-simulation games.

Multi-agent RL methods such as MADDPG [19] and value decomposition networks [20] provide starting points, as does work on emergent compositional communication [21]. The key challenge is designing reward structures that incentivize persona-*consistent* social patterns rather than converging to a single globally optimal interaction style irrespective of individual personality.

C. Long-Horizon Memory and Identity Persistence

A believable NPC should remember a grudge from ten play sessions ago. Current PCSP has no episodic memory beyond the persona embedding and within-episode trajectory. Connecting persona-conditioned behavior to long-horizon memory architectures—transformer-based context [11], retrieval-augmented episodic memory [2], or world models [22]—is an open problem.

The persona-conditioned shared policy framework naturally separates *who this NPC is* (persona embedding, static) from *what has happened to this NPC* (episodic memory, dynamic). Developing architectures that condition behavior on both, without expensive per-step LLM calls, is a key challenge.

D. Richer Worlds and Engine Integration

Mini-Inzoi's $6{\times}6$ grid with 4 agents is a tractable starting point. Scaling to commercial life-simulation density—larger worlds, more agents, richer action spaces, continuous physics, asynchronous event streams—requires rethinking both the environment interface and the policy architecture.

Melting Pot [23] provides a suite of multi-agent social scenarios at larger scale and is a natural next evaluation target. Beyond simulation, integrating trained policies with commercial game engines (Unreal Engine, Unity) as callable policy modules raises practical integration challenges—latency stacking, state serialization, engine-side randomness—that have received little academic attention.

E. Human Evaluation for Persona Fidelity

The ultimate judge of NPC believability is the player. Automatic metrics (trajectory identification accuracy, behavioral KL, Spearman ρ) are useful proxies for ablations but cannot capture whether a player experiences an NPC as authentically "introverted" or "conscientious."

Designing psychometrically valid human evaluation protocols for NPC persona fidelity is non-trivial. Existing personality assessment instruments (Big Five questionnaires [13]) were designed for real humans. Adapting methods from personality computing [24] and human-agent interaction to NPC personas—with controlled observer studies, inter-rater agreement ($\alpha \geq 0.6$), and adequate sample sizes ($n \geq 100$)—is an important open methodological task.

F. Authoring Tools for Game Designers

For this research direction to reach production, it must interface with designers' workflows. Designers author NPC personalities through prose, reference personality frameworks, and iterate based on playtester feedback. A persona-conditioned shared policy system should provide:

- An interface for writing and previewing persona descriptions.
- Rapid feedback on the predicted behavioral effect of a persona change, without requiring a full retraining cycle.
- Diagnostic tools that explain *why* an NPC is behaving as it does in terms the designer can act on.
- Control over the precision-recall trade-off between persona fidelity and task performance.

None of these tools exist in current RL-based NPC literature. Addressing them connects the technical research agenda to design practice.

VI. Evaluation Agenda

Progress on the research agenda requires standardized evaluation protocols. We propose the following minimum set of metrics for persona-conditioned NPC research, beyond task reward.

a) Persona identification accuracy.: Given a trajectory, can an observer (automated or human) identify which persona generated it? We recommend reporting both in-distribution (training personas) and zero-shot (held-out personas) k-NN accuracy. This measures how strongly behavior reflects the conditioning signal.

b) Behavioral diversity.: Mean pairwise KL divergence between action distributions conditioned on different persona embeddings. Low values indicate behavioral collapse regardless of task reward.

c) Semantic-behavioral alignment.: Spearman ρ between persona embedding cosine distances and pairwise behavioral KL divergences. This measures whether the semantic structure of persona space is faithfully reflected in behavior space—the key criterion for zero-shot generalization. High ρ with low KL variance indicates alignment without meaningful diversity; both should be reported together.

d) Inference latency.: GPU ms/step at batch size 1, with p95/p99 percentiles. The practical constraint for real-time engine integration is typically $<$16 ms/frame.

e) Human-rated believability.: Observer ratings of persona consistency and character naturalness, aggregated with inter-rater agreement (Krippendorff's α). This is the ground-truth criterion for production deployment and should not be deferred indefinitely.

We further advocate for a dedicated *persona-conditioned NPC benchmark*—perhaps an adaptation of Melting Pot [23] with curated personality profiles and standardized evaluation protocols—to enable reproducible comparison across future methods.

VII. Limitations and Scope

PCSP's current limitations are significant and should calibrate how its results are interpreted.

Minimal environments. Both Mini-Inzoi variants (6×6 / 4 agents and 12×12 / 16 agents) are far removed from commercial life-simulation complexity. While the v2 experiment shows that key architectural properties hold at larger grid scale, continuous physics, longer time horizons, and richer action semantics remain untested.

Synthetic personas. All 300 personas are algorithmically generated from Big Five archetypes and occupation templates. Whether the method works with designer-authored personas—with all their nuance, inconsistency, and cultural specificity—is untested.

No human evaluation. Zero-shot identification accuracy and Spearman ρ are automated proxies. We have not yet collected player judgments of persona fidelity. This is a necessary step before drawing conclusions about perceived NPC believability.

No engine integration. The policy runs in a Python simulation environment. Integration challenges with commercial game engines remain unknown.

These limitations are not arguments against the research direction—they are precisely the open problems that motivate §V. We present them explicitly to guard against overclaiming.

VIII. Conclusion

Life simulation games create a scaling challenge that the game AI field has not yet systematically addressed: how to give hundreds to thousands of NPCs distinct, consistent, and controllable personalities without proportional authoring or inference cost. We have argued that *persona-conditioned shared policies*—a single RL policy conditioned on frozen LLM persona embeddings—represent a promising architectural direction satisfying the four axes that matter for practical deployment, and we have presented PCSP as an early proof-of-concept supporting this argument.

The central message is not that PCSP solves NPC personalization. The central message is: *there exists a region of design space—combining language semantics, shared policies, and behavioral regularization—where persona consistency, zero-shot controllability, and real-time inference can coexist*. Exploring that region systematically is a concrete and tractable research program. We hope this paper motivates the community to take it up.

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